An Exploration of Earthquake Magnitudes: Alert Levels and Distributions

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DATA 612

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Introduction

Dataset contains information on 782 earthquakes with magnitudes greater than 6 from September 1995 to September 2023.

<u>Research Questions and Objectives:</u>

What is the relationship between magnitude and alert level?

What can different magnitude distributions tell us about earthquake trends?

Photo: https://www.nytimes.com/2024/01/01/world/asia/japan-earthquake.html

Methodology

My data was sourced from Kaggle

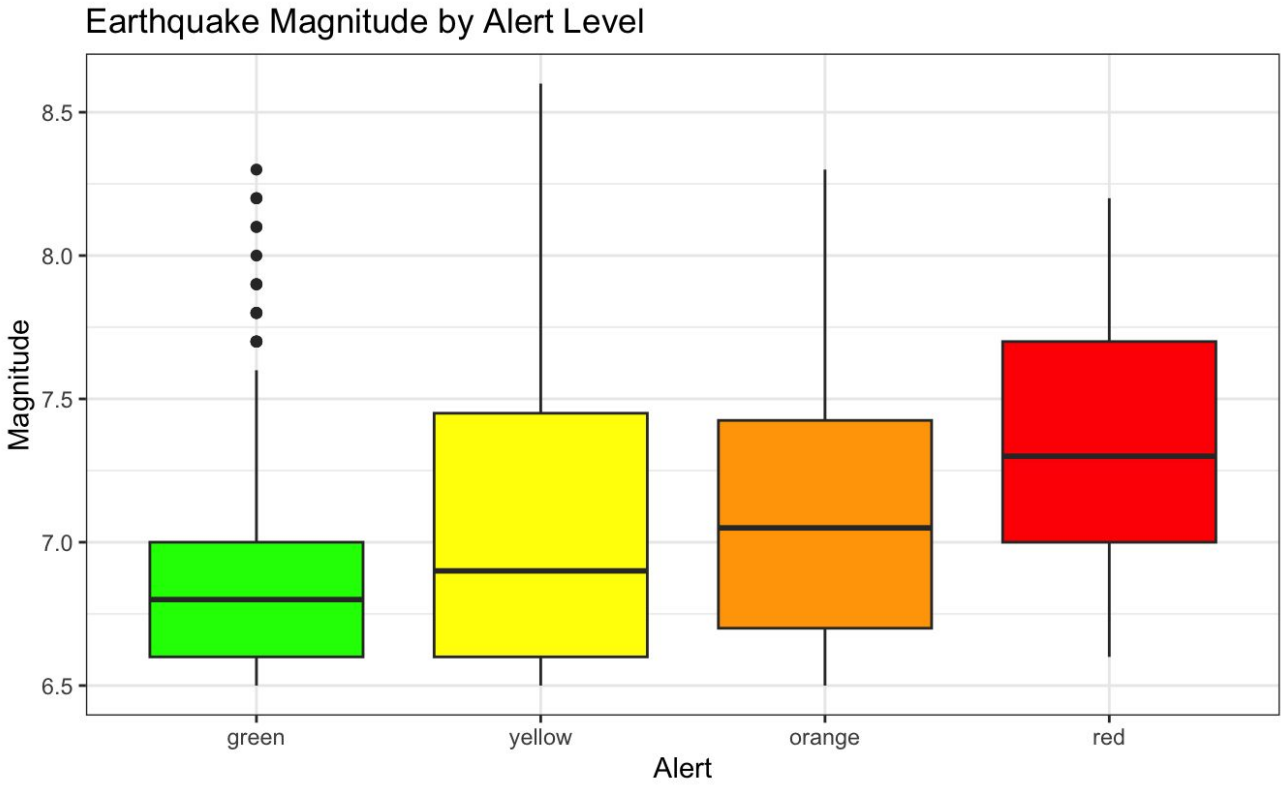
Analysis Techniques

- Exploratory and final data modeling with ggplot
- Data filtering with drop_na and mutate
- Data analysis with group_by and summarise dyplr verbs

Chirag Chauhan. (2023). *Earthquake Dataset* [Dataset]. Kaggle. Accessed 30 November 2025, from https://www.kaggle.com/datasets/warcoder/earthquake-dataset

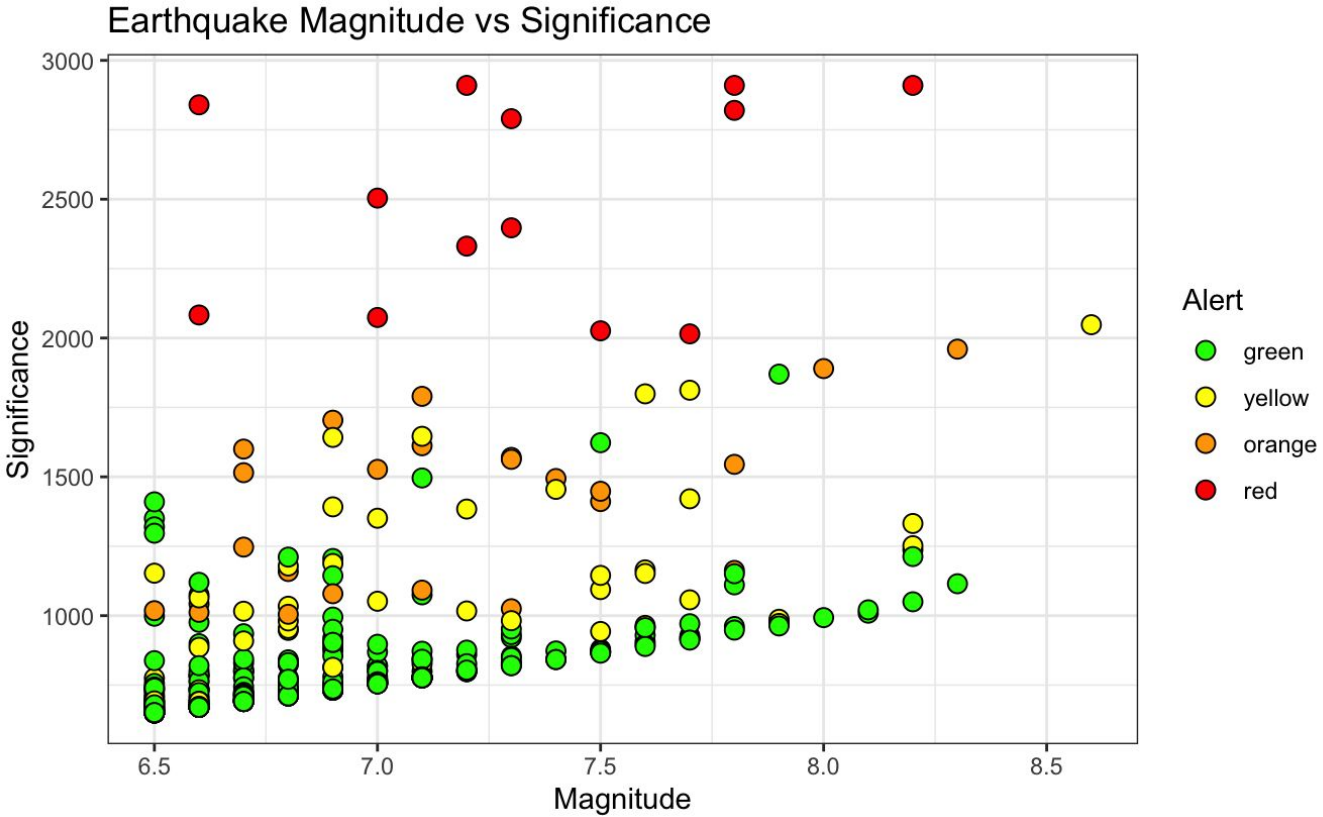
Positive correlation between earthquake magnitude and alert level.

- Interesting outliers in green alert level and large ranges in yellow and orange.



Positive correlation between magnitude and significance with alert levels increasing with significance.

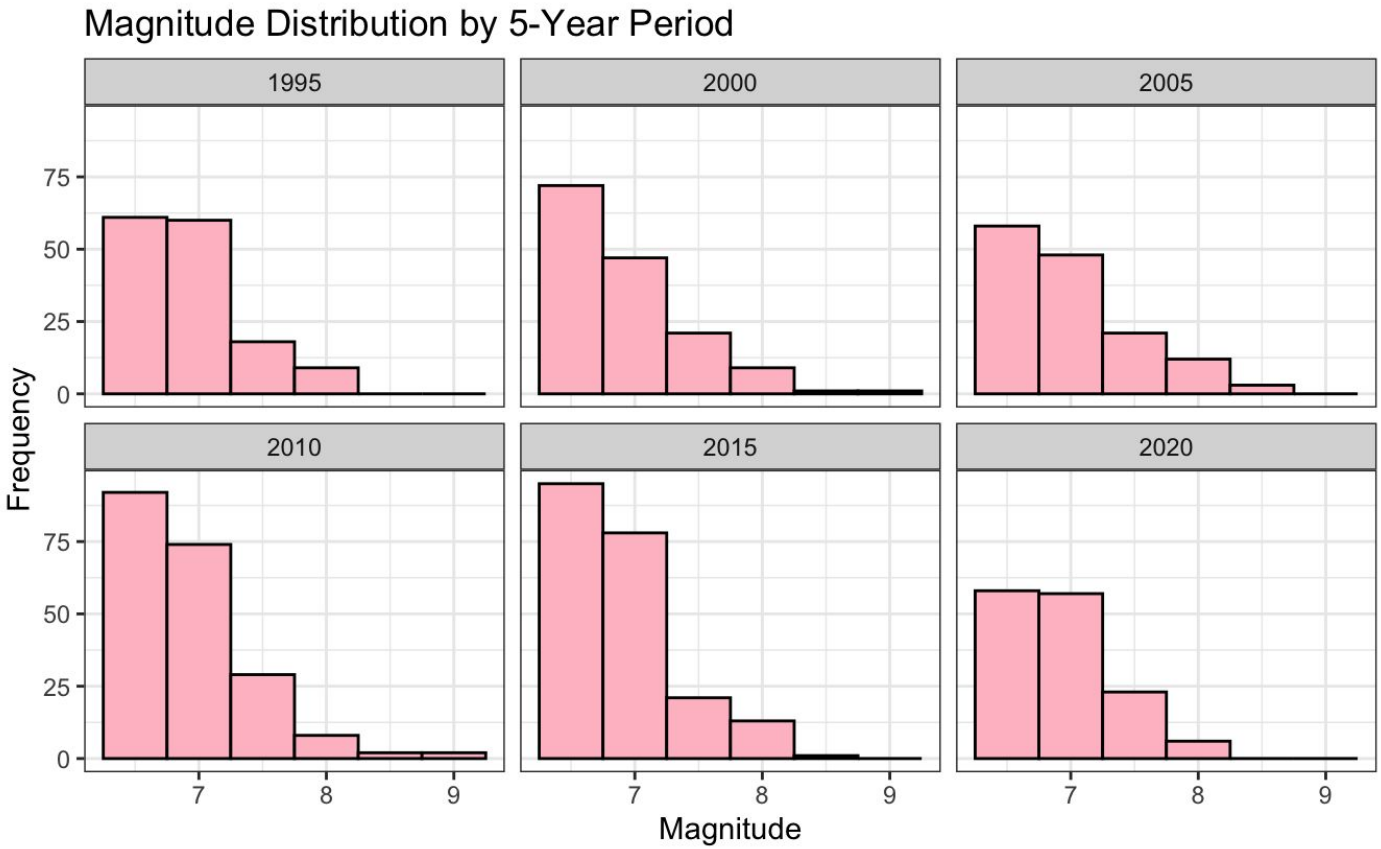
- Significance is calculated by magnitude, felt reports, and estimated impact
- Different alert levels occur at all magnitudes



Magnitude distribution remains similar throughout five year periods.

Number of earthquakes

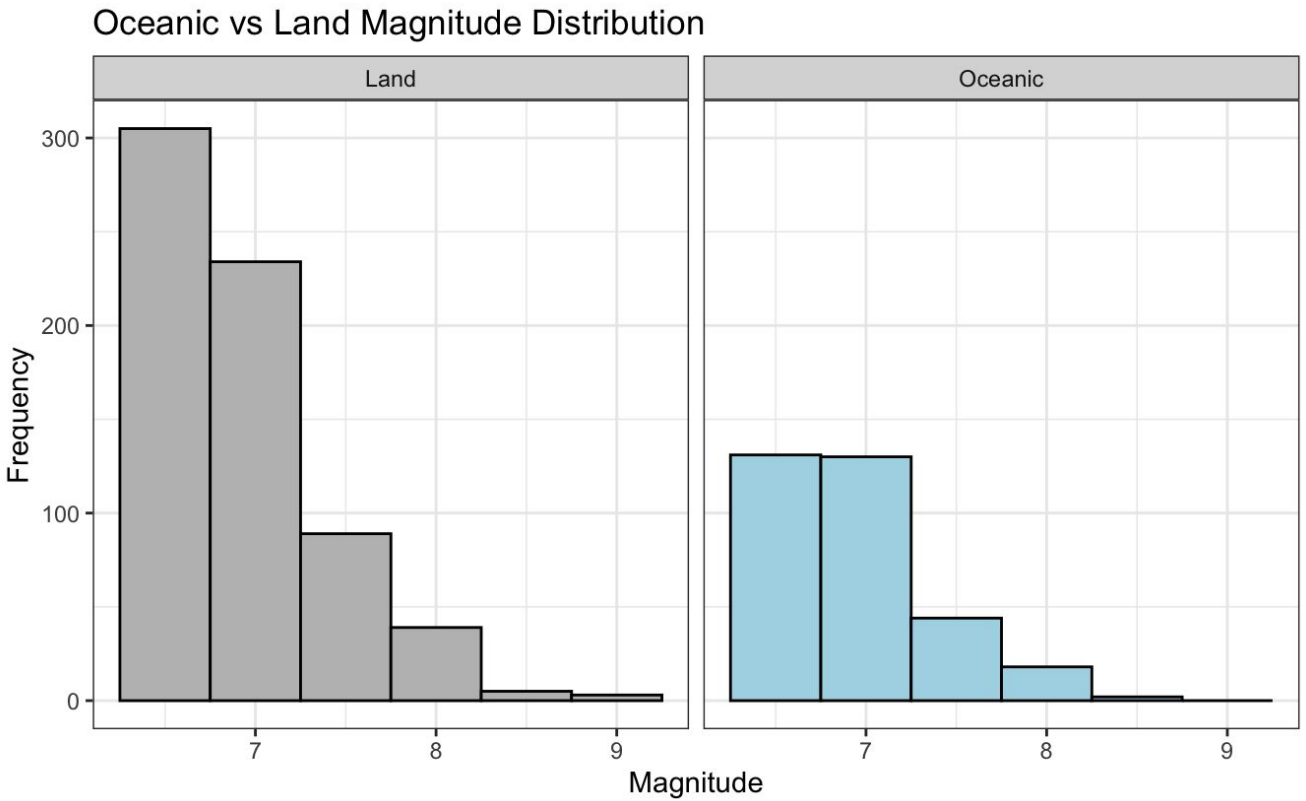
1995 - 1999	148
2000 - 2004	151
2005 - 2009	142
2010 - 2014	207
2015 - 2019	208
2020 - 2023	144



More than twice as many earthquakes occurred on land than in the ocean, with higher magnitudes on land.

Number of earthquakes

Land	675
Oceanic	325



Conclusions and Recommendations

Alert levels increase with significance as well as with magnitude. Green alert has a large number of outliers; these outliers may have been underestimated alerts.

Earthquake distributions have remained similar with an increase in earthquakes around 2010. More earthquakes occur on land than in the ocean.

<u>Further Study</u>

- Refining the analysis of alert levels by looking at how the alert level was determined.
- Looking at population near the earthquake location with significance, magnitude, and alerts.